

Advanced Electroweak Kinematics: Mathematical Formulation of Z and W Boson Mass Extractions

Sapun

Beck College of Sciences and Mathematics
Arkansas State University

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Abstract

This dossier provides a rigorous theoretical deconstruction of the mathematical models utilized in the ORIAS Monte Carlo engine to extract the gauge boson masses of the electroweak interaction (Z^0 and $W^\pm$) using CMS Run2012B data. It details the invariant mass kinematics, the C^1 continuity of the Double-Sided Crystal Ball function, the kinematic derivation of the Jacobian edge, and the Q^2 scaling of quantum chromodynamic hadronic recoil.

Contents

1	Introduction	2
2	Part I: Z Boson Kinematics and Statistical Modeling	2
2.1	Special Relativity and Invariant Mass	2
2.2	Quantum Mechanics vs. Detector Reality	2
2.3	The Double-Sided Crystal Ball (DSCB) Function	2
3	Part II: W Boson Single-Muon Extraction	3
3.1	The η Phantom and Geometric Acceptance	3
3.2	The Drell-Yan Contamination and Jacobian Decapitation	3
3.3	Hadronic Recoil and the 2D Matrix Forge	3

1 Dimuon Kinematics and the Z^0 Boson

1.1 Four-Vector Calculus and Invariant Mass

In the Standard Model, the Z^0 boson is highly unstable, decaying via the neutral current interaction $Z^0 \rightarrow \mu^+ \mu^-$ in approximately $\sim 3 \times 10^{-25}$ seconds. Detection necessitates the reconstruction of the parent particle's invariant mass from its decay products.

Let the four-momentum of the parent Z^0 boson be denoted as P_Z^μ , and the four-momenta of the daughter muons as P_1^μ and P_2^μ . By the conservation of four-momentum:

$$P_Z^\mu = P_1^\mu + P_2^\mu \quad (1)$$

The invariant mass M_{inv} is defined as the Minkowski inner product of the four-vector with itself, utilizing the metric signature $(+, -, -, -)$:

$$M_{\text{inv}}^2 \equiv P_Z^\mu P_{Z\mu} = (P_1^\mu + P_2^\mu)(P_{1\mu} + P_{2\mu}) \quad (2)$$

Expanding this product, and recognizing that $P_1^\mu P_{1\mu} = m_\mu^2$ (which is negligible at high energies, $m_\mu \approx 0.105$ GeV), we obtain:

$$M_{\text{inv}}^2 \approx 2P_1^\mu P_{2\mu} = 2(E_1 E_2 - \vec{p}_1 \cdot \vec{p}_2) \quad (3)$$

In experimental collider coordinates, spatial momentum is defined by transverse momentum (p_T), pseudorapidity (η), and azimuthal angle (ϕ). The exact kinematic expansion computed by the ROOT framework is:

$$M_{\text{inv}} = \sqrt{2p_{T,1}p_{T,2} [\cosh(\eta_1 - \eta_2) - \cos(\phi_1 - \phi_2)]} \quad (4)$$

This formulation is invariant under longitudinal Lorentz boosts, making it the fundamental equation for reconstructing resonance states in a hadron collider.

1.2 Quantum Resonance: The Breit-Wigner Distribution

The quantum mechanical nature of an unstable particle dictates that it does not possess a singular mass. The uncertainty principle ($\Delta E \Delta t \geq \hbar/2$) requires a mass distribution governed by the particle's decay width Γ . The cross-section for the scattering process forming the Z^0 resonance is described by the relativistic Breit-Wigner formula:

$$\sigma(E) \propto \frac{E^2 \Gamma^2}{(E^2 - M_Z^2)^2 + M_Z^2 \Gamma^2} \quad (5)$$

where M_Z is the pole mass and $\Gamma \approx 2.49$ GeV is the natural decay width.

2 Statistical Modeling: The Double-Sided Crystal Ball

2.1 Convolution of Resolution and Radiation

The physical CMS detector cannot observe the pure Breit-Wigner resonance. The finite spatial resolution of the silicon tracker and muon drift tubes introduces a Gaussian smearing. Furthermore, final-state Bremsstrahlung radiation ($\mu \rightarrow \mu\gamma$) siphons energy before detection, creating a heavy left tail. The measured distribution is mathematically a convolution of the Breit-Wigner probability density function (PDF), a Gaussian resolution PDF, and a radiation power-law PDF.

2.2 Mathematical Formulation of the DSCB

To optimize the computational extraction of the mass, the ORIAS engine utilizes the Double-Sided Crystal Ball (DSCB) function, a piecewise function engineered to stitch a Gaussian core to two independent power-law tails while maintaining C^1 continuity (a continuous first derivative).

Given a standardized random variable $t = \frac{x-\mu}{\sigma}$, the PDF is defined as:

$$f(t; \alpha_L, n_L, \alpha_R, n_R) = \begin{cases} A_L \cdot (B_L - t)^{-n_L} & \text{for } t < -\alpha_L \\ \exp\left(-\frac{t^2}{2}\right) & \text{for } -\alpha_L \leq t \leq \alpha_R \\ A_R \cdot (B_R + t)^{-n_R} & \text{for } t > \alpha_R \end{cases} \quad (6)$$

2.3 Proof of C^1 Continuity

The variables A and B are not free parameters; they are strictly analytically derived to enforce boundary smoothness. To ensure the function and its derivative are continuous at the left transition point $t = -\alpha_L$:

$$\lim_{t \rightarrow -\alpha_L^-} f(t) = \lim_{t \rightarrow -\alpha_L^+} f(t) \implies A_L \cdot (B_L + \alpha_L)^{-n_L} = \exp\left(-\frac{\alpha_L^2}{2}\right) \quad (7)$$

Matching the first derivatives:

$$\left. \frac{d}{dt} [A_L (B_L - t)^{-n_L}] \right|_{t=-\alpha_L} = \left. \frac{d}{dt} \left[\exp\left(-\frac{t^2}{2}\right) \right] \right|_{t=-\alpha_L} \quad (8)$$

$$n_L A_L (B_L + \alpha_L)^{-(n_L+1)} = \alpha_L \exp\left(-\frac{\alpha_L^2}{2}\right) \quad (9)$$

Solving this system of equations yields the exact computational constants embedded in the engine:

$$B_L = \frac{n_L}{|\alpha_L|} - |\alpha_L| \quad (10)$$

$$A_L = \left(\frac{n_L}{|\alpha_L|} \right)^{n_L} \exp\left(-\frac{|\alpha_L|^2}{2}\right) \quad (11)$$

The right tail parameters A_R and B_R are derived symmetrically.

3 W-Boson Kinematics: The Jacobian Edge

3.1 Derivation of the Transverse Momentum Singularity

Unlike the Z^0 , the $W^\pm$ decays asymmetrically into a charged lepton and a neutrino ($W \rightarrow \mu\nu$). Because the neutrino escapes undetected, invariant mass cannot be calculated. We must analyze the p_T spectrum of the single muon.

In the rest frame of the W boson, the muon is emitted at an angle θ^* relative to the beam axis. By conservation of momentum, the muon's momentum magnitude is $|p^*| \approx M_W/2$. The transverse momentum is therefore:

$$p_T = \frac{M_W}{2} \sin \theta^* \quad (12)$$

Assuming the W boson is unpolarized, the decay is isotropic in solid angle, meaning the distribution of $\cos \theta^*$ is uniform:

$$\frac{dN}{d \cos \theta^*} = \text{constant} \quad (13)$$

To find the distribution of transverse momentum, we apply the standard Jacobian transformation of probabilities:

$$\frac{dN}{dp_T} = \frac{dN}{d \cos \theta^*} \left| \frac{d \cos \theta^*}{dp_T} \right| \quad (14)$$

Using $\cos \theta^* = \sqrt{1 - \sin^2 \theta^*} = \sqrt{1 - \left(\frac{2p_T}{M_W}\right)^2}$, we differentiate with respect to p_T :

$$\left| \frac{d \cos \theta^*}{dp_T} \right| = \frac{4p_T/M_W^2}{\sqrt{1 - (2p_T/M_W)^2}} \quad (15)$$

Substituting this back into the distribution yields the pure, unsmeared Jacobian spectrum:

$$\frac{dN}{dp_T} \propto \frac{p_T}{\sqrt{(M_W/2)^2 - p_T^2}} \quad (16)$$

This equation reveals a profound kinematic truth: as $p_T \rightarrow M_W/2$, the denominator approaches zero, and the distribution creates an integrable singularity (a massive spike) that drops to absolutely zero for $p_T > M_W/2$. This sheer mathematical cliff is the Jacobian Edge.

3.2 Transverse Smearing and Initial State Radiation

In physical reality, the W boson is not at rest; it is given a transverse kick (q_T) by the emission of Initial State Radiation (gluons) from the annihilating quarks. This q_T smears the theoretical singularity into a smooth, finite curve.

The magnitude of this hadronic recoil scales with the energy of the hard scatter, denoted as Q^2 . For gauge boson production, $Q^2 \approx M^2$. Thus, the perturbative QCD emission probability scales linearly to the first order:

$$\langle q_T \rangle_W \approx \langle q_T \rangle_Z \times \left(\frac{M_W}{M_Z} \right) \quad (17)$$

However, the 2D Matrix Forge developed in ORIAS proves that relying solely on this analytical Q^2 scaling underestimates the single-lepton recoil. Because the single-muon analysis lacks an invariant mass cut (which acts as a filter against complex hadronic spray), the effective ISR must absorb higher-order soft gluon resummation and underlying event noise, yielding an empirically derived effective ISR of 19.4 GeV to correctly un-smear the Jacobian cliff and reveal the 80.319 GeV mass.